

◆ 1. Your Spark job is running slowly. What do you do?

✓ Answer:

To troubleshoot slow Spark jobs, I take the following steps:

- **Check the execution plan** using `.explain()` or the Spark UI to locate expensive stages or shuffles.
 - **Look for data skew** — uneven partitions can make some tasks much longer than others.
 - **Use broadcast() joins** if one of the datasets is small enough to avoid shuffling.
 - **Repartition the DataFrame** with `.repartition()` to ensure more even data distribution.
 - **Cache reusable data** using `.cache()` or `.persist()` when the same DataFrame is used multiple times.
 - Monitor using Spark UI for executor/task level metrics.
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◆ 2. How do you handle a join that is taking a long time?

✓ Answer:

When a join operation is slow, I:

- **Inspect the sizes of both datasets.** If one is small (<10MB), I apply a `broadcast()` join.
 - **Check for data skew**, especially if a join key has highly repetitive values. In such cases, I implement a **salting technique** to distribute the skewed key.
 - **Review the query plan** using `.explain()` to ensure Spark uses optimal join strategies.
 - **Ensure proper partitioning** before the join to reduce shuffle.
 - **Avoid nested or chained joins** in one transformation if possible.
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◆ 3. You need to calculate a new column based on existing columns. What do you do?

✓ Answer:

To derive a new column from existing ones in PySpark:

python

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```
df = df.withColumn("total_price", df["price"] * df["quantity"])
```

- This operation is **lazy**, meaning it won't execute until an action like `.show()` or `.write()` is called.
 - You can chain multiple transformations using `withColumn()` if needed.
 - Ensure data types are compatible (e.g., don't multiply string fields without casting).
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◆ 4. Your job failed with an Out Of Memory (OOM) error. What steps would you take?

✓ Answer:

- **Reduce partition sizes** using `.repartition()` or `.coalesce()` to control memory usage per executor.
 - **Avoid using `.collect()`** or heavy `.groupBy()` on large datasets, as they bring large volumes of data into memory.
 - **Use `.persist(StorageLevel.DISK_ONLY)`** instead of keeping data in memory if memory is limited.
 - **Tune Spark configuration**, especially:
 - `spark.executor.memory`
 - `spark.executor.cores`
 - Garbage collection options (`spark.memory.fraction`, etc.)
 - Check if wide transformations can be avoided or broken down.
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◆ 5. How do you efficiently read a 10GB+ CSV file in Spark?

✓ Answer:

- Avoid `inferSchema=True` for large files as it scans many rows. Instead, **define a schema manually**.
- After reading, **use `.repartition()`** to improve parallelism and avoid skew.
- If the file is used repeatedly, **cache it** with `.cache()`.
- Consider **converting it to Parquet** format for faster, compressed, and columnar access:

python

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```
df.write.parquet("path/output")
```

◆ 6. You use the same DataFrame multiple times in your pipeline. What would you do?

✅ Answer:

- **Persist the DataFrame** after the first expensive transformation:

python

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```
df = df.persist() # or .cache()
```

- This avoids recomputing the same DataFrame each time it's used, which saves time and resources.
- Use `.unpersist()` when the DataFrame is no longer needed to free up memory.
- Choose the right **storage level** (`MEMORY_ONLY`, `MEMORY_AND_DISK`, etc.) based on resource availability.

6. Compare data lake storage vs blob storage. When would you use one over the other?

✅ Answer:

Feature	Blob Storage	Data Lake Storage (ADLS Gen2)
Hierarchical Namespace	❌ No	✅ Yes (folders, directories)
Optimized for	Unstructured object storage	Big data analytics & high throughput workloads
Security	Basic (limited to container-level)	Advanced (POSIX permissions, RBAC, ACLs)
Performance	Suitable for general-purpose use	Optimized for analytics engines (like Spark, Hive)
Integration	General storage (images, docs, backups)	Tight integration with Azure analytics tools

📌 When to use:

- Use **Blob** for general file storage.

- Use **ADLS Gen2** for analytics workloads, hierarchical file systems, and scenarios requiring fine-grained access control and scalable parallel reads/writes.
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7. Walk me through how you'd design an end-to-end ETL pipeline from an on-premise database to Azure Databricks.

✅ **Answer:**

♦ **ETL Pipeline Design (Step-by-Step):**

1. Extract:

- Use **Self-hosted IR** in **Azure Data Factory (ADF)** to connect to on-premise SQL Server, Oracle, or other DB.
- Use ADF **Copy Activity** to pull data.

2. Land (Stage):

- Store extracted data as CSV/Parquet in **ADLS Gen2** (Bronze Layer of Data Lake).

3. Transform:

- Trigger an **ADF pipeline** to run a **Databricks notebook** using Databricks activity.
- In the notebook, use **PySpark** to clean, transform, deduplicate, join data.

4. Load:

- Write transformed data to curated zones in **Delta format** (Silver/Gold Layer) in ADLS.
- Optionally, load aggregated data into **Synapse or Power BI** datasets for visualization.

✚ Add **monitoring** via ADF alerts or logs.

8. In Python, how do you extract unique values from a dictionary? Provide a code snippet.

✅ **Answer:**

If you want to extract unique values from a dictionary:

python

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```
my_dict = {  
    'a': 10,  
    'b': 20,  
    'c': 10,  
    'd': 30  
}
```

```
unique_values = set(my_dict.values())  
print(unique_values) # Output: {10, 20, 30}
```

✦ `set()` removes duplicates from dictionary `.values()`.

9. How would you execute one notebook from another in Databricks?

✓ **Answer:**

You can use `dbutils.notebook.run()`:

python

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In Notebook A

```
dbutils.notebook.run("/path/to/NotebookB", timeout_seconds=60, arguments={"input":  
"value"})
```

✦ Parameters:

- **Path:** Relative or absolute path to target notebook
- **Timeout:** In seconds
- **Arguments:** Passed as dictionary, accessed in the target notebook using `dbutils.widgets.get("key")`

✦ This is helpful for orchestrating modular pipelines.

10. Write Python code to sum amount per customer_id given a DataFrame or list of records.

✅ **Answer:**

python

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```
import pandas as pd
```

```
data = [  
    {'customer_id': 1, 'amount': 100},  
    {'customer_id': 2, 'amount': 200},  
    {'customer_id': 1, 'amount': 50},  
    {'customer_id': 2, 'amount': 150}  
]
```

```
df = pd.DataFrame(data)
```

```
result = df.groupby('customer_id')['amount'].sum().reset_index()  
print(result)
```

📌 **Output:**

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```
customer_id amount  
0          1   150  
1          2   350
```

Alternatively, in PySpark:

python

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```
from pyspark.sql.functions import sum
```

```
df.groupBy("customer_id").agg(sum("amount").alias("total_amount")).show()
```

11. Given a Boolean matrix, count the number of islands (connected components). How would you solve it?

✅ **Answer:**

Use **Depth-First Search (DFS)** or **Breadth-First Search (BFS)**.

python

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```
def num_islands(grid):  
    if not grid:  
        return 0  
  
    def dfs(r, c):  
        if r < 0 or r >= len(grid) or c < 0 or c >= len(grid[0]) or grid[r][c] != 1:  
            return  
        grid[r][c] = -1 # mark as visited  
        dfs(r+1, c)  
        dfs(r-1, c)  
        dfs(r, c+1)  
        dfs(r, c-1)  
  
    count = 0  
    for r in range(len(grid)):  
        for c in range(len(grid[0])):  
            if grid[r][c] == 1:  
                dfs(r, c)  
                count += 1  
    return count
```

Example usage:

```
matrix = [  
    [1, 1, 0],  
    [0, 1, 0],  
    [0, 0, 1]  
]  
  
print(num_islands(matrix)) # Output: 2
```

📌 An "island" is a group of adjacent 1s (connected horizontally or vertically).
Mark each visited cell to avoid counting it again.

12. How would you create a database schema for a university: colleges, students, professors? What tables, keys, and relationships?

✅ **Answer:**

You can model this using a **relational schema** with the following tables:

📘 **Tables and Keys:**

1. **College**

- college_id (PK)
- college_name
- location

2. **Department**

- department_id (PK)
- department_name
- college_id (FK to College)

3. **Professor**

- professor_id (PK)
- name
- email
- department_id (FK to Department)

4. Student

- student_id (PK)
- name
- email
- enrollment_year
- college_id (FK to College)

5. Course

- course_id (PK)
- course_name
- department_id (FK to Department)

6. Enrollment

- student_id (FK to Student)
- course_id (FK to Course)
- semester
- grade
- *(Composite PK: student_id + course_id + semester)*

7. Teaching

- professor_id (FK to Professor)
- course_id (FK to Course)
- semester

📌 This ensures **1:N** (College → Students), **M:N** (Student ↔ Courses), and **M:N** (Professor ↔ Courses) relationships.

13. Design an optimal schema to store event logs (like clicks/swipes) for high-velocity web traffic.

✅ **Answer:**

For a high-volume, write-heavy log system:

🗄️ **EventLog Table (Wide, Denormalized, Partitioned)**

- event_id (UUID or BIGINT, PK)
- user_id (FK)
- event_type (e.g., click, swipe)
- event_time (timestamp)
- device_type (mobile, desktop)
- location (optional - city/country)
- session_id
- page_url
- metadata (JSON or MAP<string, string> — for extensibility)

 Schema Design Tips:

- Store in **Delta Lake / Parquet** for compression and performance.
- Partition by **event_date** or **event_type** for efficient querying.
- Add **Z-ordering** on frequently filtered columns (e.g., user_id or session_id).

 Avoid joins during ingestion; keep schema flat.

14. Given messy sales data, walk us through cleaning, transformation, and how you'd design reporting tables for business analytics.

 **Answer:**

 Cleaning & Transformation Steps:

1. Remove Duplicates:

- Use dropDuplicates() in PySpark or ROW_NUMBER() in SQL to keep the latest record.

2. Handle Nulls:

- Fill missing region or salesperson using business logic or default values.
- Drop rows with missing order_id or amount.

3. Data Type Fixing:

- Ensure dates are parsed correctly.
- Convert amount to Decimal or Double.

4. **Standardize Values:**

- Convert categories to lowercase/title case.
- Format phone numbers or addresses.

Reporting Table Design:

1. **Fact_Sales:**

- order_id (PK)
- customer_id
- product_id
- sales_date
- amount
- discount
- region_id
- salesperson_id

2. **Dim_Date, Dim_Customer, Dim_Product, Dim_Region, Dim_Salesperson**

 Use **star schema** for BI tools like Power BI or Tableau.

15. Describe how you'd optimize a slow-performing JOIN query joining large tables. What indexing or partitioning strategies would you use?

 **Answer:**

Optimization Strategies:

1. **Broadcast Join:**

- If one table is small (<10MB), use broadcast() to avoid shuffle.

2. **Partition Pruning:**

- Ensure tables are partitioned on join/filter columns.
- Example: sales_data partitioned by region or sales_date.

3. **Clustered Index / Z-ordering:**

- Use Z-ordering (Databricks) or clustered index on frequent filter/join columns.

4. **Avoid Data Skew:**

- If one key dominates (e.g., `region_id=1` in 90% rows), use **salting**.

5. **Repartitioning:**

- Repartition data before joining to ensure even distribution across tasks.

6. **Filter Early:**

- Apply WHERE conditions before joining to reduce row count.

7. **Statistics & Caching:**

- Update table stats so the optimizer chooses the right plan.
- Cache common lookups if reused.

📌 Always start with `.explain()` to check if joins are shuffled or optimized.